



503A & 503B Compounded Rx

Product Information · Nature Classification · Side Effect & Contraindication Reference

NATURAL

Identical or derived from naturally occurring compounds

SEMI-SYN

Structurally modified analog of a natural compound

SYNTHETIC

Fully laboratory-synthesized; no direct natural counterpart

■ **Clinical Reference Only** — Side effects and contraindications list the most commonly reported. This is not an exhaustive safety profile. Individual responses vary. Always conduct a full patient assessment before prescribing.

503A — Traditional Patient-Specific

Requires a valid individual patient prescription. Regulated by state pharmacy boards. Must comply with USP <795>/<797> standards.

• Patient-specific Rx required • State pharmacy board oversight • Custom dose & formulation • No FDA batch approval

503B — FDA-Registered Outsourcing

Federally registered under FDA oversight. May produce large batches for healthcare facilities without individual Rx. Operates under CGMP standards.

• FDA-registered & inspected • CGMP manufacturing standards • Can supply facilities without Rx • Consistent batch quality controls

GLP-1 & Metabolic Agents

Semaglutide and Tirzepatide are fully synthetic incretin mimetics. Retatrutide is a novel triple agonist (GLP-1/GIP/glucagon). Most side effects are GI-related and dose-dependent.

Product	Conc	Vial	Nature	Common Side Effects	Contraindications	Clinical Notes
Semaglutide + Glycine (2.5mg)	2.5mg/5mg/mL	1 mL	SYNTHETIC	Nausea, vomiting, diarrhea, constipation, injection site reaction	• Personal/family Hx MTC • MEN2 • Pancreatitis Hx • Pregnancy • Hypersensitivity to semaglutide • Severe renal impairment (caution)	Starter dose; weekly SC. Glycine is a natural amino acid
Semaglutide + Glycine (5mg)	2.5mg/5mg/mL	2 mL	SYNTHETIC	Nausea, fatigue, reduced appetite, GI discomfort	• Personal/family Hx MTC • MEN2 • Pancreatitis Hx • Pregnancy • Hypersensitivity to semaglutide	Standard maintenance dose
Semaglutide + Glycine (12.5mg)	2.5mg/5mg/mL	5 mL	SYNTHETIC	GI upset, pancreatitis risk (rare), headache	• Personal/family Hx MTC • MEN2 • Pancreatitis Hx • Pregnancy • Gallbladder disease (caution)	Higher-dose / longer supply
Semaglutide + B12 (2.5mg)	1.25mg/0.5mg/mL	2 mL	SYNTHETIC	Nausea, injection site pain; B12 well tolerated	• Personal/family Hx MTC • MEN2 • Leber's optic neuropathy (B12 CI) • Pregnancy	B12 supports energy & tolerability
Semaglutide + B12 (5mg)	2.5mg/0.5mg/mL	2 mL	SYNTHETIC	GI symptoms, decreased appetite, mild dizziness	• Personal/family Hx MTC • MEN2 • Pancreatitis Hx • Pregnancy	Common titration step
Semaglutide + B12 (12.5mg)	3.125mg/0.5mg/mL	4 mL	SYNTHETIC	Nausea, vomiting, gallbladder issues (long-term)	• Personal/family Hx MTC • MEN2 • Gallbladder/biliary disease • Pregnancy	Extended supply

Product	Conc	Vial	Nature	Common Side Effects	Contraindications	Clinical Notes
Tirzepatide (18mg)	9mg/mL	2 mL	SYNTHETIC	<i>Nausea, diarrhea, vomiting, decreased appetite, fatigue</i>	• Personal/family Hx MTC • MEN2 • Pancreatitis Hx • Type 1 DM (not indicated) • Pregnancy	Dual GIP/GLP-1; potent metabolic
Tirzepatide (36mg)	18mg/mL	2 mL	SYNTHETIC	<i>GI upset, injection site reactions, hypoglycemia risk</i>	• Personal/family Hx MTC • MEN2 • Concurrent sulfonylurea/insulin (hypoglycemia) • Pregnancy	Mid-range maintenance
Tirzepatide (72mg)	18mg/mL	4 mL	SYNTHETIC	<i>Nausea, constipation, pancreatitis risk (rare), hair loss</i>	• Personal/family Hx MTC • MEN2 • Gallbladder disease • Pancreatitis Hx • Pregnancy	High-dose / 4-week supply
Tirzepatide + B12 (17mg)	8.5mg/0.5mg/mL	2 mL	SYNTHETIC	<i>GI symptoms, injection site irritation</i>	• Personal/family Hx MTC • MEN2 • Leber's optic neuropathy (B12 CI) • Pregnancy	Enhanced energy support
Tirzepatide + B12 (34mg)	17mg/0.5mg/mL	2 mL	SYNTHETIC	<i>Nausea, diarrhea, reduced appetite</i>	• Personal/family Hx MTC • MEN2 • Pancreatitis Hx • Pregnancy	Mid-dose with B12
Tirzepatide + B12 (68mg)	17mg/0.5mg/mL	4 mL	SYNTHETIC	<i>GI effects, gallbladder disease risk (long-term)</i>	• Personal/family Hx MTC • MEN2 • Gallbladder/biliary disease • Pregnancy	High-dose long-acting
Retatrutide (40mg)	13.3mg/mL	3 mL pen	SYNTHETIC	<i>Nausea, vomiting, diarrhea, constipation, fatigue; rare thyroid risk</i>	• Personal/family Hx MTC • MEN2 • Pancreatitis Hx • Pregnancy • Investigational — limited safety data	Triple agonist GLP-1/GIP/glucagon; once weekly SC

Sublingual Formulations						
Needle-free delivery. GI side effects may be milder than injectable forms due to lower peak plasma concentrations.						
Product	Strength	Vol	Nature	Common Side Effects	Contraindications	Benefit Highlight
Sublingual Tirzepatide + NAD+	5–17.5mg/100mg	8 mL	SYNTHETIC	<i>Nausea, oral mucosal irritation, mild flushing (NAD+)</i>	• Personal/family Hx MTC • MEN2 • Active peptic ulcer (oral mucosal CI) • Pregnancy	Synthetic GLP-1 + bioidentical NAD+ (6 strengths)
Sublingual Semaglutide	2–10mg	8 mL	SYNTHETIC	<i>Nausea, taste disturbance, dry mouth, mild GI upset</i>	• Personal/family Hx MTC • MEN2 • Oral mucosal lesions (impairs absorption) • Pregnancy	Needle-free GLP-1; 5 strengths

Product	Strength	Vol	Nature	Common Side Effects	Contraindications	Benefit Highlight
Sublingual Tirzepatide	5–30mg	8 mL	SYNTHETIC	Nausea, reduced appetite, oral irritation, headache	• Personal/family Hx MTC • MEN2 • Pancreatitis Hx • Pregnancy	Needle-free dual agonist; 6 strengths

Single Peptide Agents						
Peptides range from bioidentical to fully synthetic. Side effects are generally mild; most commonly injection site reactions and transient GI or neurological symptoms.						
Peptide	Concentration	Nature	Common Side Effects	Contraindications	Primary Action	
AOD-9604	1.2mg/mL · 5mL	SEMI-SYN	Injection site redness, nausea (rare), headache	• Active malignancy • Hypersensitivity to GH fragments • Pregnancy/lactation	Modified GH fragment; fat lipolysis; no IGF-1 effects	
BPC-157	3mg/mL · 5mL	SEMI-SYN	Mild nausea, dizziness, transient fatigue	• Active malignancy (growth-promoting; caution) • Pregnancy/lactation • Hypersensitivity	Gut & tissue repair; anti-inflammatory; tendon healing	
DSIP	1mg/mL · 5mL	NATURAL	Vivid dreams, mild headache, excessive drowsiness; inject 30–60 min before bed	• CNS depressant co-administration (additive sedation) • Hypotension • Pregnancy/lactation	Natural neuropeptide; sleep architecture & stress regulation	
EPITHALON	2mg/mL · 5mL	SYNTHETIC	Injection site reactions, transient fatigue, vivid dreams	• Active malignancy (telomerase activation — theoretical risk) • Pregnancy/lactation • Autoimmune disease (caution)	Telomerase activation; anti-aging; pineal regulation	
GHK-Cu	10mg/mL · 5mL	NATURAL	Injection site burning/stinging; mild redness (remove from fridge 15 min prior)	• Wilson's disease • Active cancer (copper may stimulate angiogenesis) • Copper hypersensitivity • Pregnancy/lactation	Copper peptide; collagen synthesis; wound & skin healing	
GONADORELIN	1mg/mL · 5mL	NATURAL	Hot flashes, headache, mood changes, testicular ache	• Hormone-sensitive malignancy (prostate, breast) • Pituitary adenoma • OHSS risk • Pregnancy (unless medically indicated)	Bioidentical GnRH; LH/FSH support; TRT adjunct	
IGF-LR3	200mcg/mL · 5mL	SEMI-SYN	Hypoglycemia, headache, joint pain, fluid retention	• Active malignancy • Diabetic retinopathy • Hypoglycemia-prone patients • Concurrent insulin • Pregnancy/lactation	Long-arg3 analog; muscle growth; anabolic signaling	
Kisspeptin	1mg/mL · 5mL	NATURAL	Flushing, mild GnRH-related hormonal shifts	• Hormone-sensitive cancers • Active PCOS with LH hypersecretion • Precocious puberty • Pregnancy	Endogenous neuropeptide; reproductive hormone regulation	
MOTS-C	2mg/mL · 5mL	NATURAL	Injection site reactions, mild GI upset (rare); 8–12 wk on / 4–6 wk off	• Mitochondrial disease (unknown interaction) • Insulin-dependent DM (monitor glucose) • Pregnancy/lactation	Mitochondrial peptide; metabolic flexibility; exercise performance	

Peptide	Concentration	Nature	Common Side Effects	Contraindications	Primary Action
NAD+	100mg/mL · 10mL	NATURAL	Flushing, nausea, headache, chest tightness (rate-dependent with IV)	• Active malignancy (NAD+ may fuel tumor metabolism — discuss) • IV: cardiac arrhythmia (slow infusion required) • Hypersensitivity	Bioidentical coenzyme; cellular energy & DNA repair
PT-141 (Bremelanotide)	2mg/mL · 5mL	SEMI-SYN	Nausea, flushing, headache, transient BP elevation; max 8 doses/month	• Uncontrolled hypertension • Cardiovascular disease / MACE history • Concurrent antihypertensives (BP drop) • Pregnancy	Melanocortin analog; CNS-mediated libido enhancement
Sermorelin	3mg/mL · 5mL	SEMI-SYN	Injection site pain, flushing, dizziness, headache	• Active malignancy • Hypothyroidism (untreated) • Pituitary tumors • Pregnancy/lactation	Synthetic GHRH analog (1–29); stimulates GH release
Tesamorelin	3mg/mL · 5mL	SEMI-SYN	Injection site reactions, fluid retention, joint pain	• Active malignancy • Pituitary tumors • Disrupted hypothalamic-pituitary axis • Pregnancy • Hypersensitivity to mannitol	FDA-approved GHRH; visceral fat reduction; IGF-1 stimulation
Thymosin A-1	1mg/mL · 5mL	NATURAL	Generally well tolerated; mild injection site reactions	• Organ transplant recipients (may counteract immunosuppression) • Autoimmune disease (stimulates immune activity) • Pregnancy/lactation	Immune modulation; anti-viral; autoimmune support
Lipo-B	50/50/25/1mg/mL	SYNTHETIC	Injection site pain, urinary odor, mild GI effects	• Sulfa allergy (methionine — caution) • Renal/hepatic insufficiency • Pregnancy/lactation	MIC + B12; lipotropic; liver detox support
Glutathione	200mg/mL · 10mL	NATURAL	Skin lightening (frequent use), mild GI, cramping	• Active chemotherapy (antioxidant may interfere) • Hypersensitivity	Master antioxidant; detoxification; skin brightening

Prefilled Injectable Pen Blends						
Curated multi-peptide blends in 3mL prefilled pens. 10 clicks = 0.1mL. Standard: 10 clicks SC nightly, 5 days on / 2 days off. Store refrigerated.						
Blend	Composition	Format	Nature	Common Side Effects	Contraindications	Clinical Notes
Wolverine Blend	BPC-157 10mg + TB-500 10mg	3mL pen	SEMI-SYN	Injection site reaction, mild fatigue, dizziness, headache	• Active malignancy • Autoimmune conditions (TB-500 immune activity) • Pregnancy/lactation	Soft tissue repair, recovery & mobility; ~333mcg BPC-157 + 333mcg TB-500/dose
Recovery Blend	BPC-157 10mg + TB-500 10mg + GHK-Cu 50mg + KPVP 10mg	3mL pen	SEMI-SYN	Injection site irritation, flushing (GHK-Cu), mild fatigue; allergic reaction warning on label	• Wilson's disease (GHK-Cu) • Active malignancy • Copper hypersensitivity • Pregnancy/lactation	Tissue healing + inflammation reduction + collagen + whole-body recovery
Glow Blend	BPC-157 10mg + TB-500 10mg + GHK-Cu 50mg	3mL pen	SEMI-SYN	Injection site burning/stinging (GHK-Cu), mild redness, transient fatigue	• Wilson's disease (GHK-Cu) • Active cancer (copper-based) • Copper hypersensitivity • Pregnancy/lactation	Skin health, collagen production & aesthetic rejuvenation

NAD+ Injectable Pens — Cellular Energy Support

Compounded injectable NAD+ for subcutaneous self-administration. Administer 2–3x/week. Rotate injection sites.

Product	Concentration	Nature	Common Side Effects	Contraindications	Clinical Notes
NAD+ 600mg	200mg/mL · 3mL pen	NATURAL	Flushing, nausea, headache, chest tightness (rate-dependent); milder SC vs IV	• Active malignancy (discuss risk/benefit) • Cardiac arrhythmia history (chest tightness monitoring) • Hypersensitivity	Standard dose pen; 50mg=25 clicks, 100mg=50 clicks; energy, cognition
NAD+ 1000mg	333mg/mL · 3mL pen	NATURAL	Flushing, nausea, headache, chest tightness (rate-dependent)	• Active malignancy (discuss risk/benefit) • Cardiac arrhythmia history • Hypersensitivity	High-dose pen; 50mg~15 clicks, 100mg~30 clicks; advanced cellular & metabolic support

Growth Hormone Support Peptides

GHRH analogs that stimulate endogenous GH release. Do not suppress natural GH production. Inject nightly 5 on/2 off.

Product	Concentration	Nature	Common Side Effects	Contraindications	Clinical Notes
Tesamorelin (GH Support Blend)	30mg / 3mL pen (1mg/dose)	SEMI-SYN	Injection site reactions, fluid retention, joint pain, headache	• Active malignancy • Pituitary tumors • Disrupted hypothalamic-pituitary axis • Pregnancy • Hypersensitivity to mannitol	GHRH analog; fat metabolism, muscle preservation, tissue repair; 10 clicks = 1mg nightly
Sermorelin (GH Support Peptide)	10mg / 3mL pen (0.33mg/dose)	SEMI-SYN	Injection site pain, flushing, dizziness, headache; very well tolerated overall	• Active malignancy • Hypothyroidism (untreated) • Pituitary tumors • Pregnancy/lactation	GHRH analog (1–29); lean mass, fat loss, sleep quality, anti-aging; 10 clicks = 0.33mg

Specialty Peptide Pens

Targeted peptides for specific clinical goals including sexual health, sleep, mitochondrial optimization, and skin regeneration.

Product	Concentration	Nature	Common Side Effects	Contraindications	Clinical Notes
PT-141 (Bremelanotide)	10mg / 3mL pen (0.33mg/10 clicks)	SEMI-SYN	Nausea, flushing, headache, elevated BP; max 1 dose/24h, 8 doses/month	• Uncontrolled hypertension • Cardiovascular disease / MACE history • Concurrent antihypertensives • Pregnancy	Melanocortin agonist; CNS sexual arousal; inject 45–120 min before activity
DSIP	5mg / 3mL pen (0.17mg/10 clicks)	NATURAL	Vivid dreams, mild headache, excessive drowsiness, mild hypotension	• CNS depressant use (additive sedation) • Hypotension / orthostatic hypotension • Pregnancy/lactation	Natural sleep neuropeptide; 100–300mcg nightly 30–60 min before bed; 4–8 wk cycles
MOTS-C	20mg / 3mL pen (0.67mg/10 clicks)	NATURAL	Injection site reactions, mild GI upset (rare); generally well tolerated	• Insulin-dependent DM (monitor glucose — AMPK activation) • Mitochondrial disease (insufficient data) • Pregnancy/lactation	Mitochondrial peptide; AMPK activation; 3–10mg/week; 8–12 wks on, 4–6 wks off

Product	Concentration	Nature	Common Side Effects	Contraindications	Clinical Notes
GHK-Cu	50mg / 3mL pen (1.7mg/10 clicks)	NATURAL	Injection site burning/stinging; mild redness (remove from fridge 15 min prior)	• Wilson's disease • Active cancer • Copper hypersensitivity • Pregnancy/lactation	Copper peptide; collagen, skin elasticity, hair density, tissue repair; 5 on/2 off

Combination Peptide Protocols					
Curated multi-peptide blends for synergistic outcomes. Side effects reflect the combined profile of each component.					
Combination	Dose/mL	Nature	Common Side Effects	Contraindications	Synergistic Goal
AOD-9604 + MOTS-C	1.2/2mg	SEMI-SYN	Injection site reaction, mild nausea (rare)	• Active malignancy • Insulin-dependent DM (monitor) • Pregnancy	Fat oxidation + mitochondrial metabolism
AOD-9604 + MOTS-C + Tesamorelin	1.2/2/3mg	SEMI-SYN	Fluid retention, joint pain, injection site effects	• Active malignancy • Pituitary tumors • Pregnancy	Full body recomposition stack
BPC-157 + TB-500	3/3mg	SEMI-SYN	Mild fatigue, dizziness, transient headache	• Active malignancy • Pregnancy/lactation	Accelerated tissue & injury repair
BPC-157 + KPV + TB-500	3/3/3mg	SEMI-SYN	Nausea, mild lethargy, injection site reactions	• Active malignancy • Autoimmune flares (KPV immune-modulating) • Pregnancy	Repair + gut inflammation + healing
BPC-157 + GHK-Cu + KPV + TB-500	3/10/3/3mg	SEMI-SYN	Flushing, injection site irritation, mild fatigue	• Wilson's disease (GHK-Cu) • Active malignancy • Copper hypersensitivity • Pregnancy	Full-spectrum regenerative protocol
CJC-1295 + Ipamorelin	1.2/2mg	SYNTHETIC	Water retention, tingling, mild hunger, headache	• Active malignancy • Pituitary tumors • Diabetic retinopathy (elevated IGF-1) • Pregnancy	GH pulse optimization; body composition
Tesamorelin + Ipamorelin	3/2mg	SEMI-SYN	Fluid retention, joint pain, injection site pain	• Active malignancy • Pituitary tumors • Pregnancy • Hypersensitivity to mannitol	Visceral fat + GH secretagogue synergy
DSIP + BPC-157 + CJC-1295	1/2/2mg	SEMI-SYN	Drowsiness, headache, mild fluid retention	• CNS depressants (additive sedation — DSIP) • Active malignancy • Pituitary tumors • Pregnancy	Sleep, repair & GH — recovery stack
GHK-Cu + Epithalon	10/2mg	SEMI-SYN	Injection site reactions, transient fatigue	• Wilson's disease • Active malignancy (both components) • Copper hypersensitivity • Pregnancy	Skin rejuvenation + telomere anti-aging
SEMAX + SELANK	1/1mg	SEMI-SYN	Mild headache, nasal irritation (intranasal), fatigue	• Severe psychiatric disorders (limited data) • MAOI use (theoretical serotonergic interaction) • Pregnancy/lactation	Cognitive clarity + anxiolytic balance

Combination	Dose/mL	Nature	Common Side Effects	Contraindications	Synergistic Goal
Pinealon + PE22-28 + SELANK	2/2/2mg	SYNTHETIC	Transient headache, dizziness, drowsiness	- Severe psychiatric disorders - CNS depressants (additive) - Pregnancy/lactation - Limited human data — caution in frail patients	Neuroprotection + memory + calm focus

Oral Capsule Formulations					
Product	Dose	Nature	Common Side Effects	Contraindications	Indication / Use
BPC-157	500mcg capsule	SEMI-SYN	Mild nausea, GI discomfort (rare at oral doses)	- Active malignancy - Pregnancy/lactation	Gut healing & systemic anti-inflammatory
Ondansetron	4mg tablet	SYNTHETIC	Headache, constipation, QT prolongation (rare)	- Congenital long QT syndrome - Concurrent QT-prolonging agents - Apomorphine co-administration (severe hypotension) - Hypersensitivity to ondansetron or 5-HT3 antagonists	Anti-nausea; GLP-1 side-effect management
SLU-PP 332	250mcg capsule	SYNTHETIC	Limited human data; potential GI upset, insomnia	- Investigational — no established human safety profile - Cardiac conditions (ERR agonism; monitor) - Pregnancy/lactation	Exercise mimetic; mitochondrial biogenesis
5-Amino (5-AMP)	50mg capsule	NATURAL	Mild GI upset, fatigue at higher doses	- Insulin-dependent DM (AMPK activation; glucose monitoring) - Metformin co-use (additive AMPK; hypoglycemia risk) - Pregnancy/lactation	AMPK activator; metabolic & cellular energy
Ibutamoren (MK-677)	25mg capsule	SYNTHETIC	Increased appetite, water retention, fatigue, numbness	- Active malignancy (GH/IGF-1 elevation) - Uncontrolled DM (worsens glucose tolerance) - Congestive heart failure (fluid retention) - Pregnancy	Oral GH secretagogue; deep sleep; recovery
Dihexa	5mg capsule	SYNTHETIC	Headache, irritability, vivid dreams (limited human data)	- Investigational — very limited human safety data - Hormone-sensitive conditions (HGF pathway activation) - Pregnancy/lactation	Nootropic; synaptic formation; memory & learning
SLU-PP 332 + BAM	100mcg/15mg	SYNTHETIC	GI discomfort, potential CNS stimulation	- Investigational combo — no human safety profile - Psychiatric disorders (CNS stimulation) - Cardiac conditions - Pregnancy	Dual metabolic activator combo
Dihexa + Tesofensine	5mg/500mcg	SYNTHETIC	Elevated HR/BP, dry mouth, insomnia, nausea	- Uncontrolled hypertension - Cardiovascular disease / arrhythmia - MAOI/serotonergic agent co-use (serotonin syndrome) - Eating disorders - Pregnancy	Cognitive + appetite control synergy

Ancillary & Support Agents

Product	Concentration	Nature	Common Side Effects	Contraindications	Clinical Notes
NAD+ (1200mg)	200mg/mL · 6mL	NATURAL	Flushing, nausea, chest tightness, headache (rate-dependent)	• Active malignancy (discuss) • Cardiac arrhythmia (IV rate-dependent — slow infusion) • Hypersensitivity	IV/IM; bioidentical coenzyme; energy, detox, longevity
Glutathione (2000mg)	200mg/mL · 10mL	NATURAL	Skin lightening with frequent IV use, abdominal cramping	• Active chemotherapy (antioxidant interference with oxidative agents) • Hypersensitivity	IV push antioxidant; liver, immune, skin brightening
Methylcobalamin B12 (10mg)	1mg/mL · 10mL	NATURAL	Generally very well tolerated; rare acne, mild diarrhea	• Leber's hereditary optic neuropathy • Hypersensitivity to cobalamin or cobalt • Polycythemia vera (caution — stimulates RBC production)	Bioidentical active B12; neuro support; methylation
Sermorelin (15mg)	2.5mg/mL · 6mL	SEMI-SYN	Injection site pain, flushing, headache, fluid retention	• Active malignancy • Hypothyroidism (untreated) • Pituitary tumors • Pregnancy/lactation	Synthetic GHRH analog; anti-aging; GH stimulation

IMPORTANT NOTICE: This document is intended for licensed healthcare practitioners only. All compounded medications require a valid patient-practitioner relationship and appropriate clinical evaluation. Side effects and contraindications listed are the most commonly reported and do not constitute an exhaustive safety profile. These products have not been evaluated or approved by the FDA for the specific indications listed. 503A products require individual patient prescriptions. 503B products may be distributed to registered healthcare facilities per federal guidelines. 'Natural' classifications refer to structural origin only and do not imply greater safety or efficacy. Contraindications represent standard clinical caution categories; provider judgment required in each case. **For professional use only — not for retail distribution.**